

# WASP-12b

R-1572

Benchmark run · workflow W8-benchmark-deep · record deep-bench\_WASP-12\_190123 · created 2026-08-01T22:48:18+00:00

BENCHMARK: ACCURATE — fit consistent with published values

## Results

|                                                   |                                  |
|---------------------------------------------------|----------------------------------|
| Mid-Transit Time (Tmid)                           | 2458506.8479 +/- 0.00057 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.12 +/- 0.02                    |
| Transit depth (Rp/Rs)^2                           | 1.44 +/- 0.48 [%]                |
| Orbital Inclination (inc)                         | 76.95 +/- 2.95                   |
| Airmass coefficient 1 (a1)                        | 1.181 +/- 0.017                  |
| Airmass coefficient 2 (a2)                        | -0.1028 +/- 0.0045               |
| Scatter in the residuals of the lightcurve fit is | 1.41 %                           |
| Best Comparison Star                              | None                             |
| Optimal Aperture                                  | 4.65                             |
| Optimal Annulus                                   | 19.21                            |
| Transit Duration (day)                            | 0.102 +/- 0.017                  |

## Accuracy vs published ephemeris (ExoClock/NEA)

|                           |                                        |
|---------------------------|----------------------------------------|
| Rp/R■ fit vs published    | 0.12 ± 0.02 vs 0.1178 (deviation 0.1σ) |
| Duration fit vs published | 0.102 d vs 0.125 d (deviation 1.23σ)   |
| Mid-time O–C              | 1.1 min                                |
| Depth SNR                 | 5.5                                    |
| Residual scatter          | 1.41 %                                 |

## Light curve

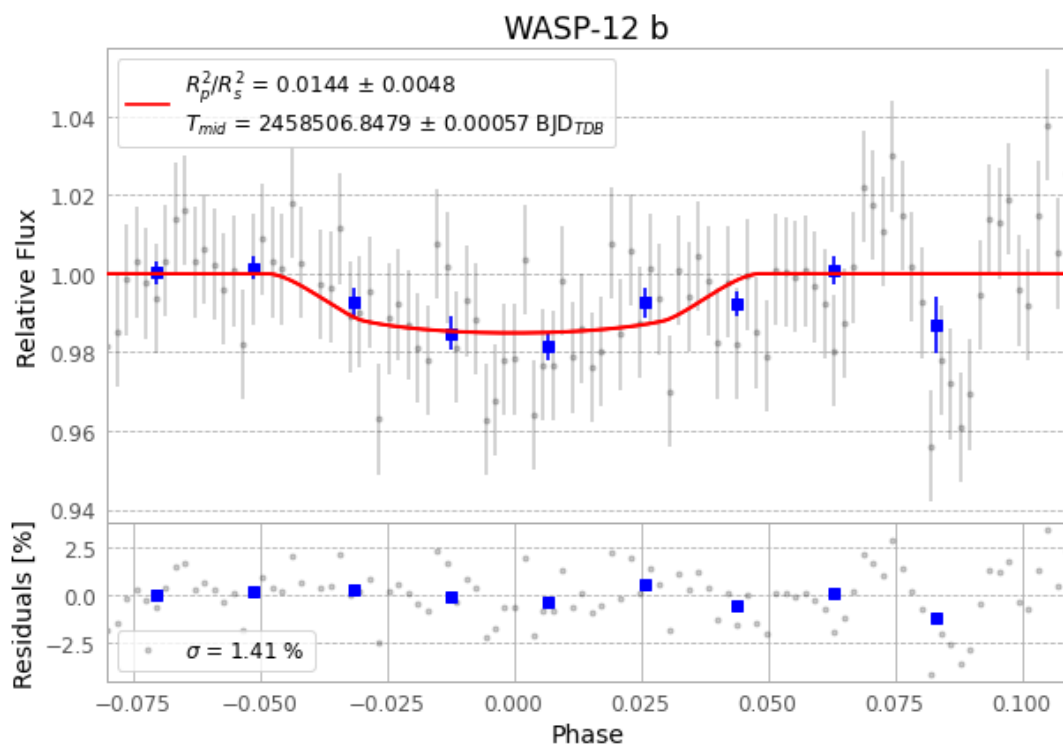


Figure 1 — FinalLightCurve\_WASP-12 b\_2019-01-22.png (SHA-256 5f401a6879fab308...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [399, 216], 3 comparison star(s). Exact configuration: parameters.inits\_json in record.json (SHA-256 cfb1e4182fa6f9fb...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit ba3ef8b

## Data provenance

100 FITS frames (66 MB), WASP-12190123060612.FITS → WASP-12190123110312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-190123.log (1cf37e61a87a...), FinalLightCurve\_WASP-12 b\_2019-01-22.png (5f401a6879fa...), FinalLightCurve\_WASP-12 b\_2019-01-22.csv (9a8edf03ad92...)

## Compute resources

Wall time 160.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

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Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 22:48Z.